

ARYAN M SHAKYA

aryanshakyaoofficial.mail@gmail.com | +91 93223 11917

[GitHub - Aryan-Shakya](#) | [Website - aryanshakya.in](#) | [LinkedIn - aryan-shakya](#)

SUMMARY

Innovative Student & Tech Enthusiast with a strong foundation in Robotics, IoT, and Web Development. Experienced in developing integrated hardware-software systems and building scalable web applications. Proven ability to lead technical initiatives, including founding a university club, completing a rural internship, and conducting technical workshops for schools. Early adopter of **vibe coding** and AI-augmented development workflows to accelerate project lifecycles.

TECHNICAL SKILLS

- **Programming Languages:** Python, JavaScript, C/C++, Java, SQL, HTML5, CSS3.
- **Robotics & Embedded Systems:** ESP32/Raspberry Pi (4,5 & Zero 2W), Teensy, Sensor Integration, Real-time Processing.
- **Tools :** Git & GitHub, VS Code, Arduino IDE, Firebase, AI Tools like - Antigravity, Gemini, Claude.
- **Professional Skills:** Leadership, Management, Communication, Outreach, Teamwork, Organization, Analysis.

EDUCATION & ACHIEVEMENTS

- **Bachelor of Technology in Computer Science and Engineering** | D. Y. Patil International University | 2028 | XII - Nav Maharashtra Vidyalaya Pimpri | School - New Millenium English Medium School
- **Emerging Talent Scholarship** – Awarded by D.Y. Patil International University (DYPIU) in the first year for academic excellence and high merit in entrance examinations.
- **Awarded 1st RUNNER UP at National Level Hackathon – OFFGRID-1.0**

EXPERIENCE

- **Group Lead & Technical Intern | Rural Internship**
 - ❖ **Organized and conducted** AI-focused workshops for local schools, successfully mentoring over **400+ students** in the foundational principles of **using AI for education and studying**.
 - ❖ **Featured in local newspapers** for community impact and leadership in bringing emerging technology to rural education systems
- **Co-Founded & President | University Robotics & IoT Club - YANTRIX**
 - ❖ **Launched** a student-led organization focusing on the integration of Cloud, AI, and Robotics systems.
 - ❖ **Spearheaded** the development of the "**University AI Reception Assistant Robot**," an automated system designed to streamline campus navigation and visitor inquiries.
- **AIML Intern | Choice Tech Labs, Banner Pune**
 - ❖ **Engineered a Retrieval-Augmented Generation (RAG)** pipeline integrating custom knowledge bases with **Large Language Models (LLMs)** to deliver highly accurate, context-aware responses.
 - ❖ Optimized **data ingestion and vector search processes**, significantly improving information retrieval efficiency and **reducing model hallucinations** across complex datasets.

ACADEMIC PROJECTS

- **Reception Desk Bot | Raspberry Pi 4, Whisper, Piper, ChromaDB, Ollama, Python**
 - ❖ **Engineered** a fully local voice pipeline integrating Whisper (STT), ChromaDB semantic search (600+ Q&A entries), and Llama3.2:3b for intelligent response generation.
 - ❖ **Deployed** the system using Piper (TTS) on a Raspberry Pi 4 to facilitate real-time interaction.
 - ❖ **Achieved** a significant reduction in student navigation time, providing instant location and faculty information.
- **Dual-Motor Stabilization Aircraft | IMU Stabilization, Flight Control, Embedded Systems (ongoing)**
 - ❖ **Designed** a specialized flight system with dual-motor propulsion and vertical transition capabilities.
 - ❖ **Implemented** IMU stabilization protocols to ensure flight balance and precision control during complex transitions.
- **Internet-Controlled RC Car | ESP8266, Arduino, Firebase, HTML/CSS/JS**
 - ❖ **Developed** a Wi-Fi enabled RC car featuring a custom web-based control interface with sub-200ms command latency.
 - ❖ **Integrated** Firebase Realtime Database to enable low-latency, remote operation of physical hardware from any browser.
- **IoT Device Monitoring Dashboard | ESP32, Firebase, JavaScript, Sensors**

- ❖ **Built** a real-time dashboard ingesting live telemetry (ultrasonic, pulse, humidity) from ESP32 microcontrollers.
- ❖ **Visualized** data trends and implemented threshold alerting systems to enable remote hardware monitoring without physical proximity.

INTERESTS

Playing cajon & guitar, Cycling, Cooking.